Abhay Sawale

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Professional Summary

Computer Science undergraduate at IIIT Kottayam (2027) with hands-on experience in software development, Python programming, REST API development, and AI-integrated applications. Experienced in building full-stack platforms, asynchronous processing pipelines, authentication systems, and database-driven applications using Python, Node.js, and modern web technologies. Interested in developing scalable software solutions and applying AI/ML techniques to real-world problems. Seeking opportunities in Software Development, Python Engineering, or AI Systems.

Technical Skills

Programming Languages:	Python, C++, C, Java, JavaScript, TypeScript
Backend Development:	FastAPI, Flask, Node.js, Express.js, REST APIs
Frontend Development:	React.js, Next.js, HTML5, CSS3
Databases:	PostgreSQL, MySQL, MongoDB
AI/ML:	LLM Integration, Prompt Engineering, Agentic AI Pipelines, Pandas, NumPy
Cloud & DevOps:	AWS S3, Docker, Git, CI/CD Fundamentals
Tools & Technologies:	Redis, BullMQ, FFmpeg, Firebase, Prisma, Postman

Projects

Animatube — AI Video Generation Platform [GitHub] *Next.js · BullMQ · Redis · FFmpeg · Groq · AWS S3*

- Built an automated AI video generation pipeline integrating LLM-based script generation, Edge TTS voice synthesis, and FFmpeg-based video assembly.
- Implemented asynchronous background processing using BullMQ and Redis to execute video generation jobs independently of the main application request flow.
- Developed real-time job-status updates using Server-Sent Events (SSE), enabling clients to receive pipeline progress without repeated polling.
- Integrated OAuth authentication, token storage, and AWS S3 asset management to support authenticated users and cloud-based video storage.

Shiv Shankar Restaurant Platform [Live Demo] [GitHub] *React · Node.js · MongoDB · Firebase · PayPal API*

- Shipped a full-stack restaurant platform handling end-to-end order, payment, and chatbot flows by architecting a modular Node.js/Express backend with controllers, middleware, and service layers.
- Enabled real-money transactions with full order traceability by integrating PayPal's REST API and persisting every transaction state to MongoDB with rollback-safe schemas.
- Reduced attack surface across all API endpoints by enforcing Helmet headers, input validation, and rate limiting — hardening the backend against common OWASP vulnerabilities.
- Implemented Firebase authentication with email verification and OAuth support, enabling secure user onboarding with zero custom auth infrastructure.

CodeForge — Online Coding Judge Platform [GitHub] *Python · FastAPI · PostgreSQL · Redis · Celery · Docker*

- Built a full-stack online coding judge with FastAPI and PostgreSQL, enabling users to solve programming problems, submit solutions, and track submission history.
- Implemented asynchronous code evaluation using Celery and Redis, processing submissions through background workers and maintaining persistent execution results.
- Integrated an isolated code execution service to evaluate solutions against test cases, handling verdicts such as Accepted, Wrong Answer, Runtime Error, and Time Limit Exceeded.
- Developed REST APIs for problem management, submissions, and user progress, with authentication, role-based access control, and protected hidden test cases.

Education

Indian Institute of Information Technology (IIIT) Kottayam 2023 – 2027
B.Tech in Computer Science Engineering

Problem-Solving Competitive Programmings || Certifications

- Solved 300+ LeetCode problems across Dynamic Programming, Arrays, Sliding Window, Trees, and Graphs.
- Regularly practice algorithmic problem-solving and participate in coding contests to strengthen data structures and system design fundamentals.
- Python Certification By Udemy